

Alain Hernandez

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WORK EXPERIENCE



Iron Galaxy Studios

Game Developer, Programming
Dec 2023 to Present

Responsible for major features in Diablo 2: Resurrected - Infernal Edition shipped release

- Gameplay sublead

Experience with legacy ports

- Ported Call of Duty: Black Ops 1, 2, and 3 social features to GDK



Meta Reality Labs

Augmented Reality Hardware/Software Prototyping
Feb 2021 to January 2023

Technical owner for prototype research AR display pipeline

- Coordinated cross-functionally with industry experts to accelerate development of novel real-time AR glasses experiences, including real-time optical distortion correction and panel uniformity.
- Designed and implemented a modular OpenGL pipeline framework adopted by internal teams that enabled SITL testing and development before hardware integration.
- Lead an initiative to introduce telemetry and error tracking in remote prototype devices leveraging Meta's existing infrastructure.



Zipline

Software Engineer
Sep 2016 to July 2020

Offline Flight Route Validation System Lead - 2018 to 2020

- Tracked air and ground risks and implemented validators that improved delivery reliability. Integrated with on-ground UI (React) workflow used by 40+ technicians to accelerate delivery site onboarding.
- Lead development of scalable architecture (AWS Batch, PostGIS) and flight route storage to improve the rate we onboarded new delivery sites (~1/week to ~25/week).

SUMMARY

Software Engineer with a decade of experience building critical systems in aerospace, AR/VR, and AAA gaming. With a proven track record as a Technical Lead, I specialize in transforming complex stakeholder requirements into scalable solutions, having led roadmaps for distributed systems at Zipline and real-time graphics pipelines at Meta.

SKILLS

Languages: Fluent in **Python, C++**, OOP; Proficient in C, Javascript

Fullstack: PostgreSQL (SQL), Flask, Nginx, uWSGI, Django, React, Tailwind, Node, Typescript

System: Linux, Windows, Perforce, Git, Mercurial, Vim, Bash

EDUCATION



University of Pennsylvania

School of Engineering and Applied Science, May 2016

B.S.E. in Computer Science | Minor in Electrical Engineering



Zipline (continued)

Software Engineer

Sep 2016 to July 2020

Technical Lead of Fulfillment System (Logistics and Inventory) - 2016 to 2018

- Lead development of Django/Postgres app, React/Redux/MobX frontend that interoperated with a local deployment (offline capability) and AWS cloud (EC2, ECS, RDS).
- Architected and ran the technical roadmap for a distributed CRM platform capable of receiving and tracking orders from hospitals and providing unit-level traceability information of 1,000+ units.
- Mentored two interns and grew the Fulfillment Tech team to three people.

Supported bring-up of two generations of Zipline's vehicles - 2016 to 2020

- Overhauled control surface calibration process of aircraft to dramatically improve time to launch (~10 min to <2 min) using April Tags + detectors that interoperated between an iPhone and embedded processors.
- Integrated new generation battery management system into HITL set-up and designed battery server protocol to transfer logs from multiple redundant logging nodes on the aircraft to a cloud server while the batteries charged and new app code was flashed onto the aircraft.

Bring-up of Unmanned Traffic Management (UTM) solution - 2020

Built a React + Cesium frontend application with a Flask/PostgreSQL backend that securely and reliably provided telemetry updates of in-air fleet over cellular to regulators (Rwanda and Ghana Civil Air Authorities).

LEADERSHIP & ENTREPRENEURSHIP



Fable Coffee

Specialty Coffee Shop, Founder & Co-Owner

Jan 2023 - Present

- **Built and launched** a high-volume specialty coffee brand, including local permitting, bar workflow design, recruiting, menu design, and community-driven marketing initiatives.
- **Iterated on operational processes** including opening/closing protocols and quality assurance to ensure consistent delivery for ~150 daily customers.